FNB HACKATHON 2026

UniversityFlow

From Grade 10 to graduation.

A companion for South African learners. It checks every university's own rules, keeps the dates, and stays to the last exam.

Tumo Mogame

Tumoolo







GRADE 10 UNIVERSITY GRADUATION

Five years where learners fall through the cracks

GRADE 9 TO 10

Subject choice

Maths or Maths Literacy decides which degrees stay open.

GRADE 11

Results

Many universities make early offers on final Grade 11 marks.

GRADE 12

Apply and fund

Six universities, six ways to score, six closing dates. Then NSFAS.

FIRST YEAR

Arrive

A new city, registration, modules. Many leave after year one.

EVERY YEAR

Finish

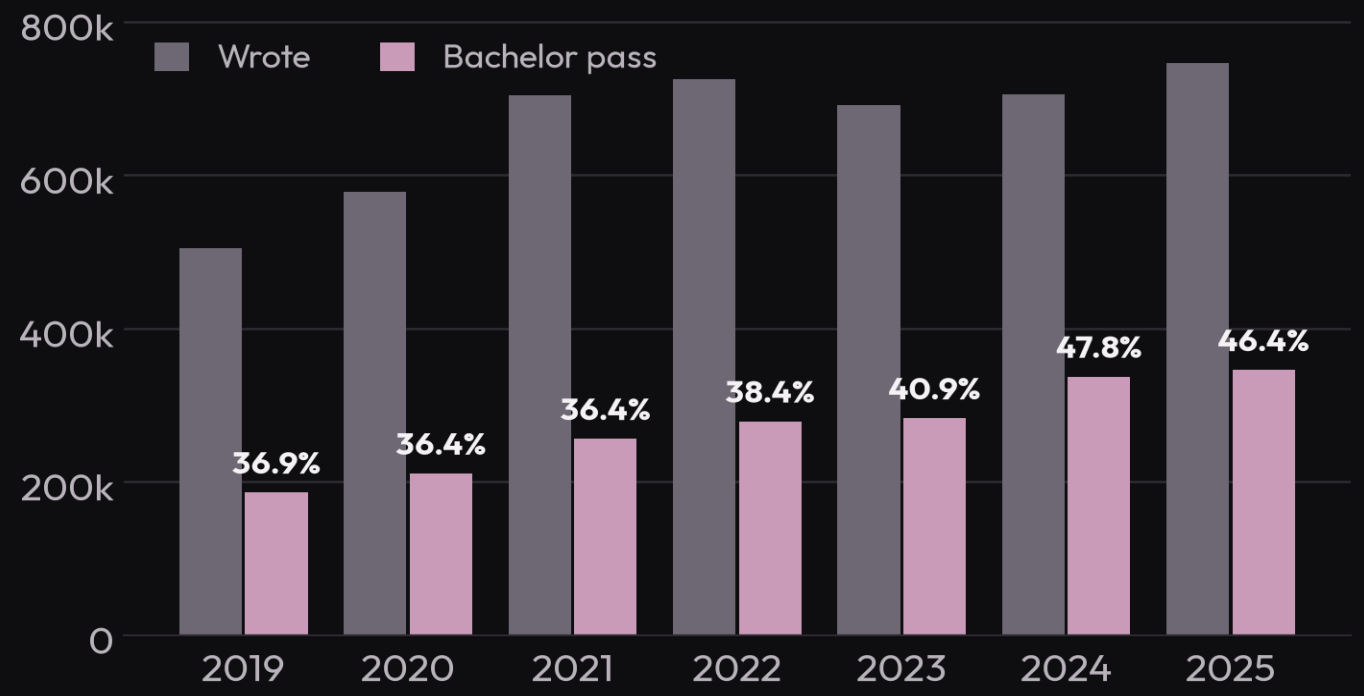
DP, subminimum, supplementary: learned the day it costs a year.

"They may not know what an APS is, which subjects close which doors, when applications close, or what NSFAS needs from them."

Quote: UniversityFlow plan, §1.1 (who it's for). The admission rules are summarised from the six universities' prospectuses (plan §2).

Under half earn a bachelor pass

2025: 746 110 full-time candidates wrote. **345 857 (46.4%)** earned a bachelor pass, the pass needed to apply for a degree.



Source: DBE, NSC Examination Reports 2021, 2024 and 2025 (full-time candidates). Labels show the bachelor-pass rate.





03 · MATHEMATICS

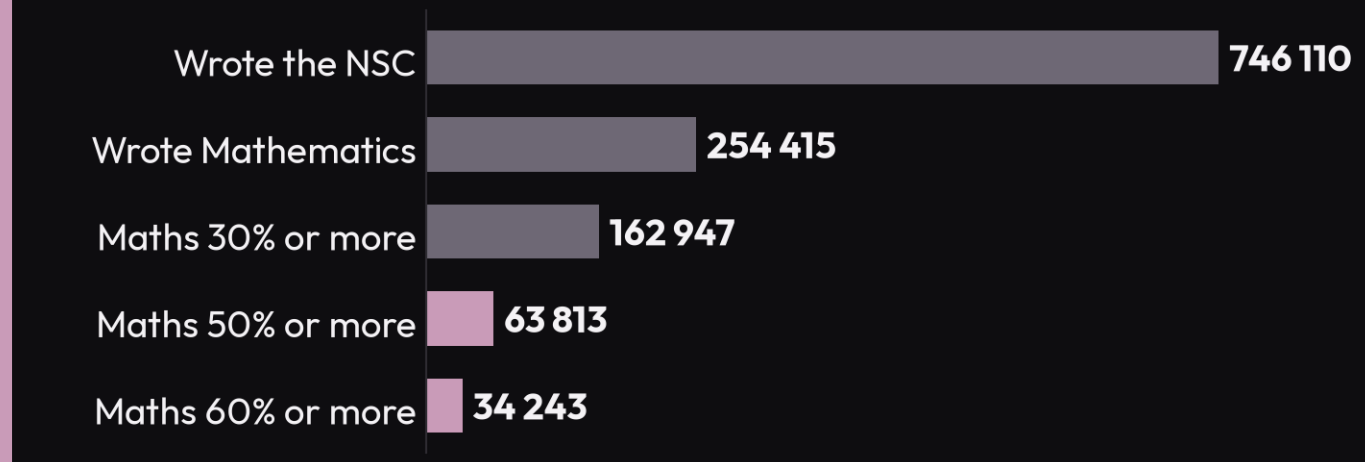
Maths closes doors early

254 415
wrote Mathematics

477 160
wrote Maths Literacy

25.1%
of Maths candidates scored 50% or more

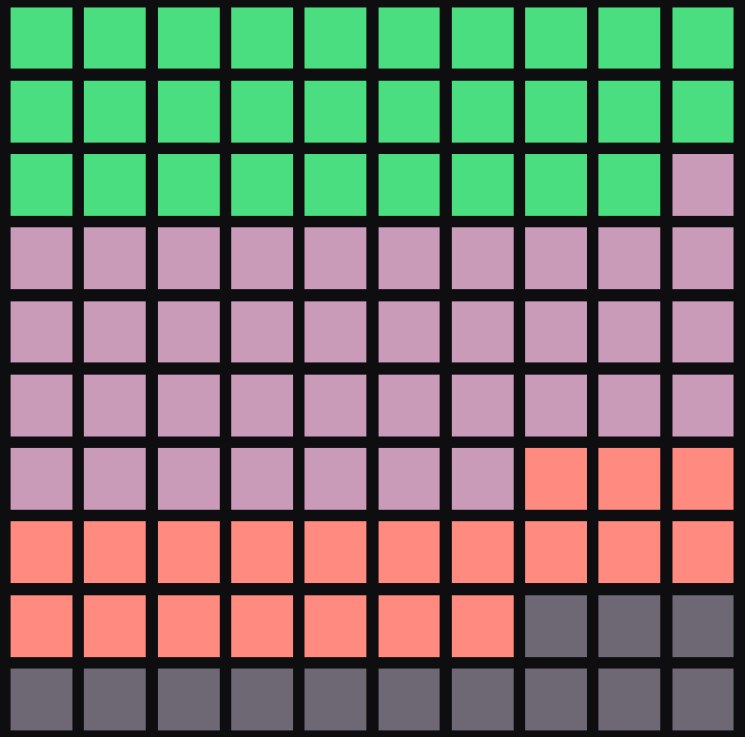
NSC 2025, full-time candidates



Source: DBE, 2025 NSC Examination Report (Tables 1, 2.1, 3, 6.3.8). Subjects are chosen before Grade 10; many science, engineering and commerce degrees ask for Mathematics.

Getting in is only half the journey

Of 100 who started a three-year degree in 2015



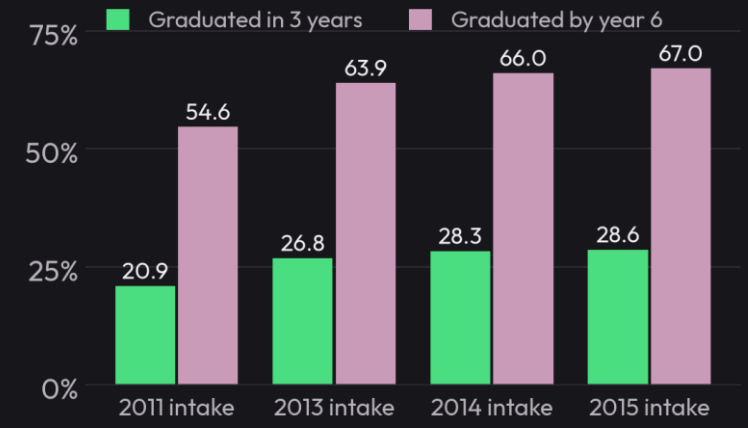
28.6%
graduated in three years

38.4%
more by year 6 (67.0% in all)

20.2%
dropped out by year 6, 13.9% of them after year one

12.8%
had neither graduated nor dropped out by year 6

Finishing on time, by intake



Better each year, but fewer than 3 in 10 finish on time.

Source: DHET, 2000 to 2019 First Time Entering Undergraduate Cohort Studies (2022), Table 28: three-year degrees, contact and distance (includes UNISA). Squares rounded; labels exact.

Funding and jobs raise the stakes

593 543

university students funded by NSFAS in 2025

R47.86 billion paid out for them that year.

R350 000

household income a year: the NSFAS limit

R600 000 for students with a disability. Applying is free.

62.8%

unemployment rate for young people aged 15 to 24

Q2 2026. 3.8 million are not in work, education or training.

UniversityFlow is built so the learner who gets in stays in, and finishes.

Sources: Minister of Higher Education, reply to Parliament NW874 (2026, via PMG); NSFAS criteria as published by universities (e.g. Rhodes); Stats SA QLFS Q2 2026.

INTRODUCING

UniversityFlow

A friend from Grade 10 to graduation.

It checks a learner's marks against each university's own rules, keeps every date, explains funding, stays close through the exams, and keeps going after the offer: arrival, modules, every year to the last exam.

Code decides. The AI explains. A person is always there.



Naledi, 16: the first in her family to aim for university

GRADE 11

Her report, read

She takes a photo of her report. Each mark is read, then confirmed by her. She sees what she is on track for, what is close, and a Plan B.

GRADE 11

A goal she can see

BCom Accounting at UJ. What I need gives the mark to aim for in each subject. Her plan map ticks off each step.

GRADE 12

Applied on time, for free

A checklist and closing date for UJ and Wits, her NSFAS documents ready, and a warning: nobody can sell her a place.

GRADE 12

Through the exams

A countdown to each paper, at most one message a day, and care first on the night she says she is not coping.

JANUARY

The first week

She accepts UJ's offer. A first-week checklist, the campus tour, and her modules checked before the last day to change them.

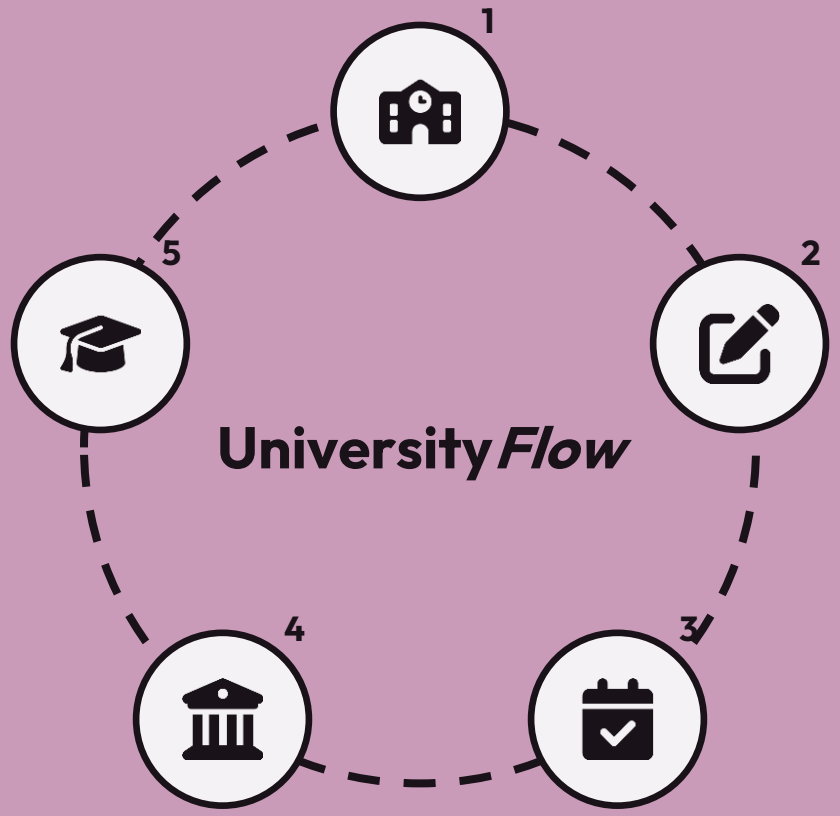
EVERY YEAR

Every module

How each module is marked, her DP, the exam mark she still needs, free books. Then the next year, to graduation.

Every step above is built and on the live demo. Naledi is made up, and so is everything she enters: try her yourself.

One companion for every stop



1 School

BUILT

Marks checked against each university's rules: qualify now, close, or Plan B. Subjects that keep doors open.

2 Apply

BUILT

A checklist and closing date per university. NSFAS: who qualifies, the documents, the free application.

3 Matric

BUILT ON STAND-INS

A goal, the exam timetable, reminders and care through the exams. Re-checks after each report.

4 Arrive

BUILT

Offers, campus, the first week, and modules checked against the published curriculum.

5 University

BUILT

Module cards, the mark calculator, free books, plan maps and My CV now. Next: a study coach.

Access to higher education: five focus areas, answered

Applications, bursaries and NSFAS	Every course scored the university's own way, a checklist and closing date each, NSFAS and ten bursaries, deadline first, scam warnings.	**Apply and fund**
Cost planning	What NSFAS pays and who qualifies, the application fee step, money safety before a deposit. Next: a cost-of-study planner.	**Money, planned**
Student wellness	Care before facts, a crisis check before any AI (red team 100%), real helplines, and Report this reply.	**Care · Crisis**
Academic persistence	Registration check, module cards, DP and the exam mark still needed, free books. Next: the study coach.	**Arrive · University years**
Connectivity and accessible learning	Light pages for cheap phones, SMS and WhatsApp, an offline calculator, Read aloud, WCAG 2.2 AA tested.	**For every learner**

Right-hand column: the slide that shows it. Focus areas as listed in the FNB Hackathon 2026 challenge "Access to higher education".

Your marks, against each university's own rules

Qualify now, close, or Plan B

Each course scored the way that university scores it: APS, points or its own formula.

What I need

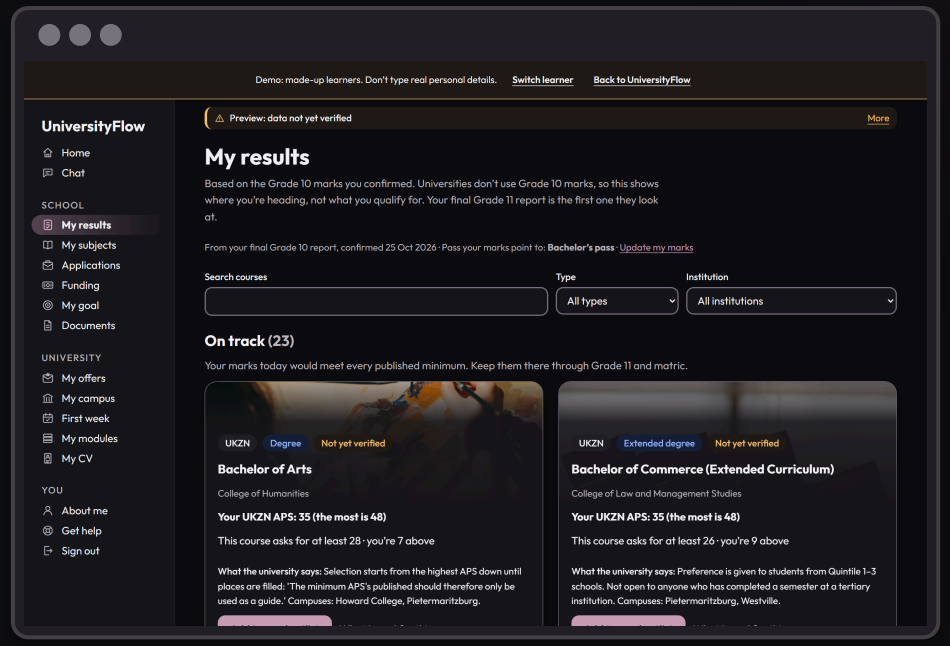
The mark to aim for in each subject, for the course the learner wants most.

Subjects that keep doors open

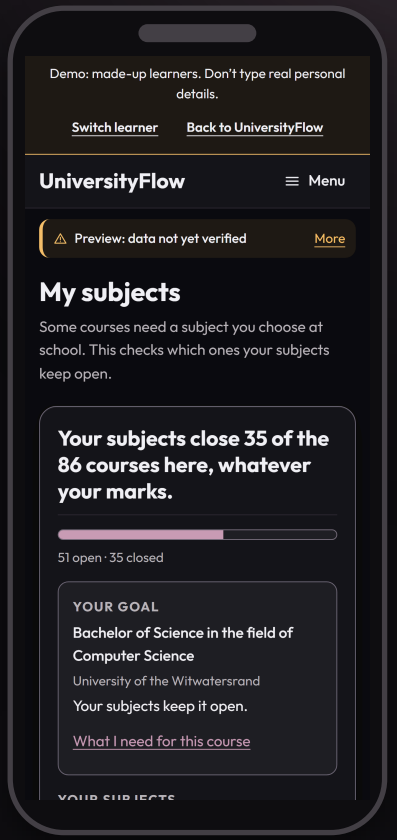
From Grade 10: which courses a subject choice opens or closes, before it's too late to change.

Only checked facts

Every requirement carries its source and year. Unchecked records are labelled and never reach real learners.



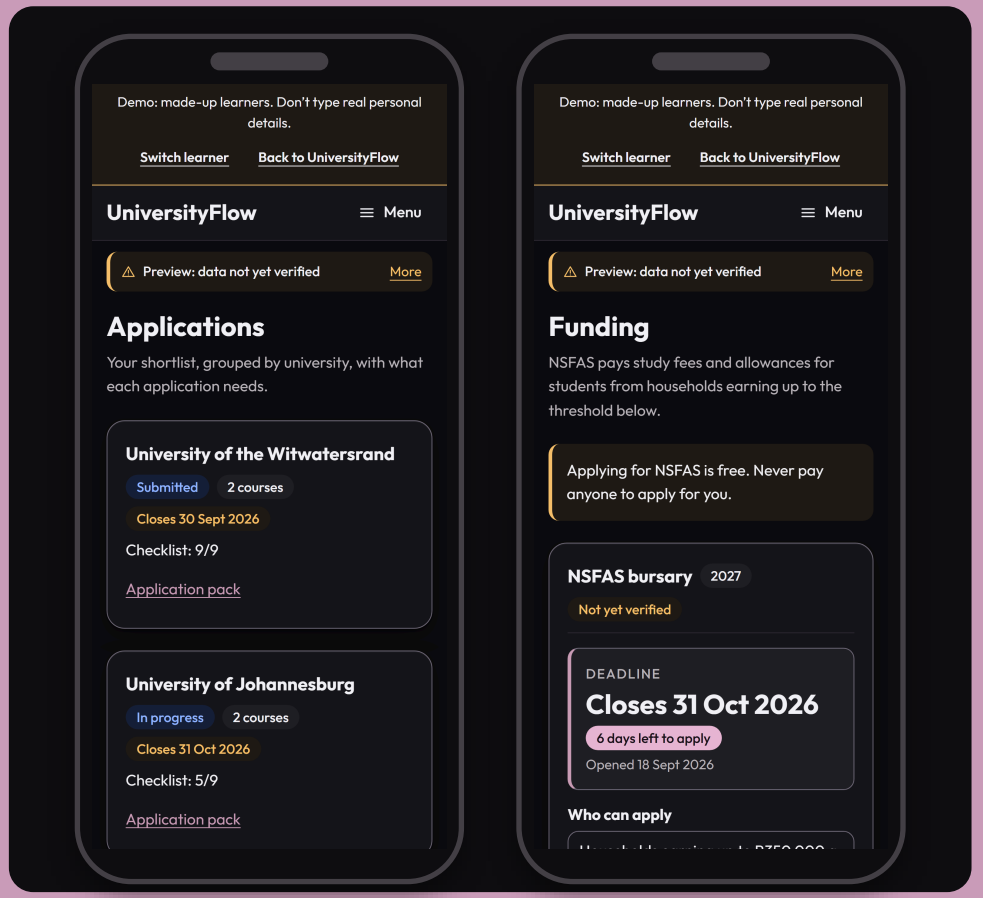
Naledi, Grade 11: My results



Thabo, Grade 10: My subjects

Today: 86 programmes at Wits, UJ, UP, UCT, TUT and UKZN, researched as drafts and awaiting a person's verification. Screens: the live judge demo, made-up learners.

Applications on time. Funding explained



Applications

BUILT

A checklist and closing date for each university, and the next step on the home page.

NSFAS

BUILT

Who can apply, the window, the documents, and keeping funding (the N+ rule). It explains; NSFAS decides.

Follow-ups

BUILT ON STAND-INS

Reminders when an application still has something to do, and before an offer's reply-by date.

Scam warnings

BUILT

Nobody can sell a place or a bursary. Applying for NSFAS is free.

Ten bursaries beside NSFAS

BUILT

Funza Lushaka, ISFAP, Thuthuka and seven more: the deadline first, who can apply, what it covers. Pre-filled applications need a funder partner.

Screens: the live judge demo (made-up learner Amahle, Grade 12). Pre-filled funding applications are gate H3 of the plan: only with a signed agreement.

Cost planning, honestly

What NSFAS pays, who qualifies

BUILT

Study fees and allowances, for households earning up to R350 000 a year (R600 000 with a disability). The documents, and how to keep funding (the N+ rule).

Application fees

BUILT

The fee step on each university's checklist: paid, or a waiver approved. Some charge nothing: UJ has no application fee.

Money safety

BUILT

Never pay anyone for a place, a bursary or an NSFAS application. A check before paying a deposit on an offer.

Ten bursaries beside NSFAS

BUILT

Each with its deadline, who can apply and what it covers, as drafts a person checks. Pre-filled applications need a funder partner.

NEXT: PROPOSED

A cost-of-study planner

For each course on the shortlist: tuition, residence, transport, books and data, against what NSFAS or a bursary pays, and the gap the family must plan for.

Built from each university's published fees, and shown only once a person has checked them. A wrong fee breaks a family's budget; we will not guess one.

Built: the Funding page (packages/content funding.ts, safety.ts), the application checklist, the offers reminders. NSFAS limits as published by universities (e.g. Rhodes); UJ: "No application fee" (its 2027 prospectus, a draft record).

Close through the exams, never in the way

TERM	A WEEK BEFORE	PAPER DAY	NEW REPORT
Up to 2 a week	**Up to 1 a day**	**Silent**	**Re-check**
Only about their own goal. Nothing to say, nothing sent.	Study reminders they asked for, never above their own cap.	Nothing from an hour before a paper to 30 minutes after.	June and September marks re-run what they qualify for.

Motivation that is theirs

The home keeps their goal and the mark each subject needs in view; the chat recalls the goal in their words.

Consent first

Opt-in per channel; a guardian's for under-18s. No messages at night. STOP works at once.

Rules as built in packages/messaging (cadence.ts); the numbers are the owner's to confirm. SMS, WhatsApp and push run on stand-ins until a provider is chosen.

UniversityFlow · SMS

UniversityFlow: this is the study reminder you asked for. Time for a little practice today. You've got this. Reply STOP to stop.

Every message is fixed, reviewed text. No name and no marks ever travel in a message: the details wait inside the app.

BUILT ON STAND-INS



13 · CARE

When a learner is struggling

Care comes before facts. The companion asks how they are before it asks about marks.

A bad result

Kind and honest: what the marks still allow, and Plan B. Never a promise it can't keep.

Exam stress

Never "don't worry". One small next step from their own timetable, and sleep before a paper.

Giving up

Their own goal, in their own words, without pressure. Another route is not a failure.

Pressure at home

Helps them find the words for family, a teacher or the school counsellor.

An AI, and says so

Asked if it's a person, it says no, and names a parent, teacher or counsellor.

Arriving alone

A lost offer or a hard first week gets care first, then real next steps.

As built: the charter and its care guidance (packages/agent, charter.ts, system-prompt.ts). UniversityFlow is not therapy and never a substitute for a counsellor, doctor or psychologist.

A crisis is answered by code, before any AI

1

A learner writes

Every message, in the chat or a plan map, is checked first.

2

A fixed check

Words of suicide, self-harm or abuse are matched by rules, not a model.

3

Coaching stops

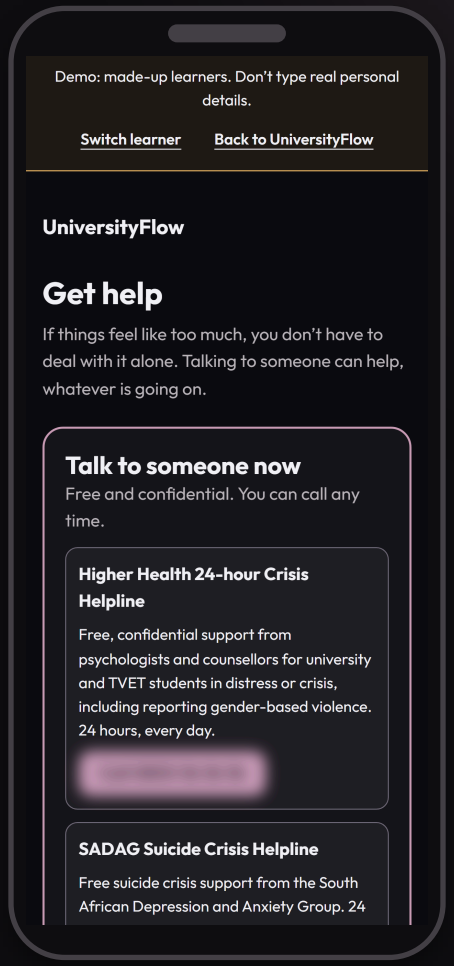
A reviewed reply and the crisis card: free helplines, then a person.

It errs towards help — A false alarm costs a few seconds; a miss could cost far more.

Tested on every change — A red-team suite of over 450 crisis cases must pass 100% for any change to the agent or its words.

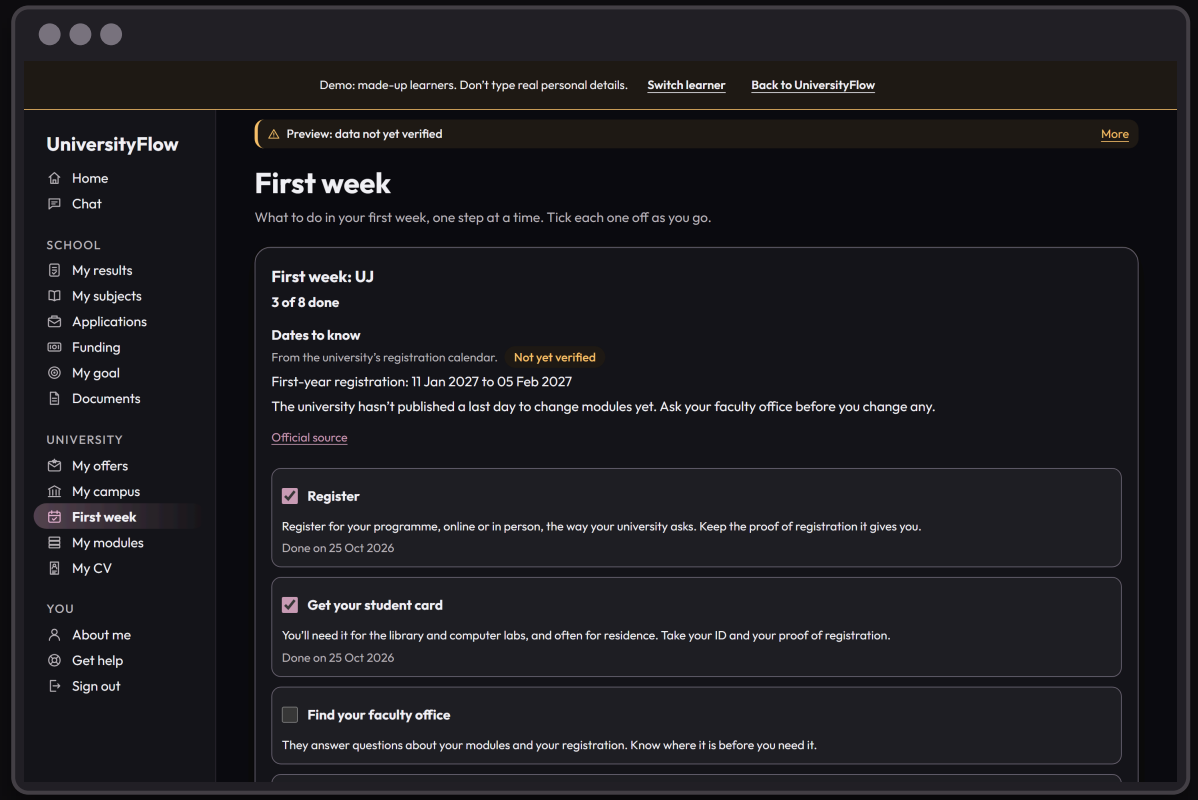
Report this reply — A button on every AI reply, so a learner can flag a bad answer.

Owed before launch, by people: each helpline number checked on the organisation's own site; crisis phrases in isiZulu, isiXhosa, Sesotho, Setswana and Afrikaans reviewed by first-language speakers; the care guidance reviewed by a clinician or Childline (A10-53).



Get help: the crisis card's helplines. Numbers blurred here until a person verifies them.

From the offer to the first lecture



Lerato, first year at UJ: First week

Offers

What kind of offer it is, what it still depends on, and the date to reply by.

Campus and first week

Where things are, walking routes, and the university's own safety advice, quoted.

Registration check

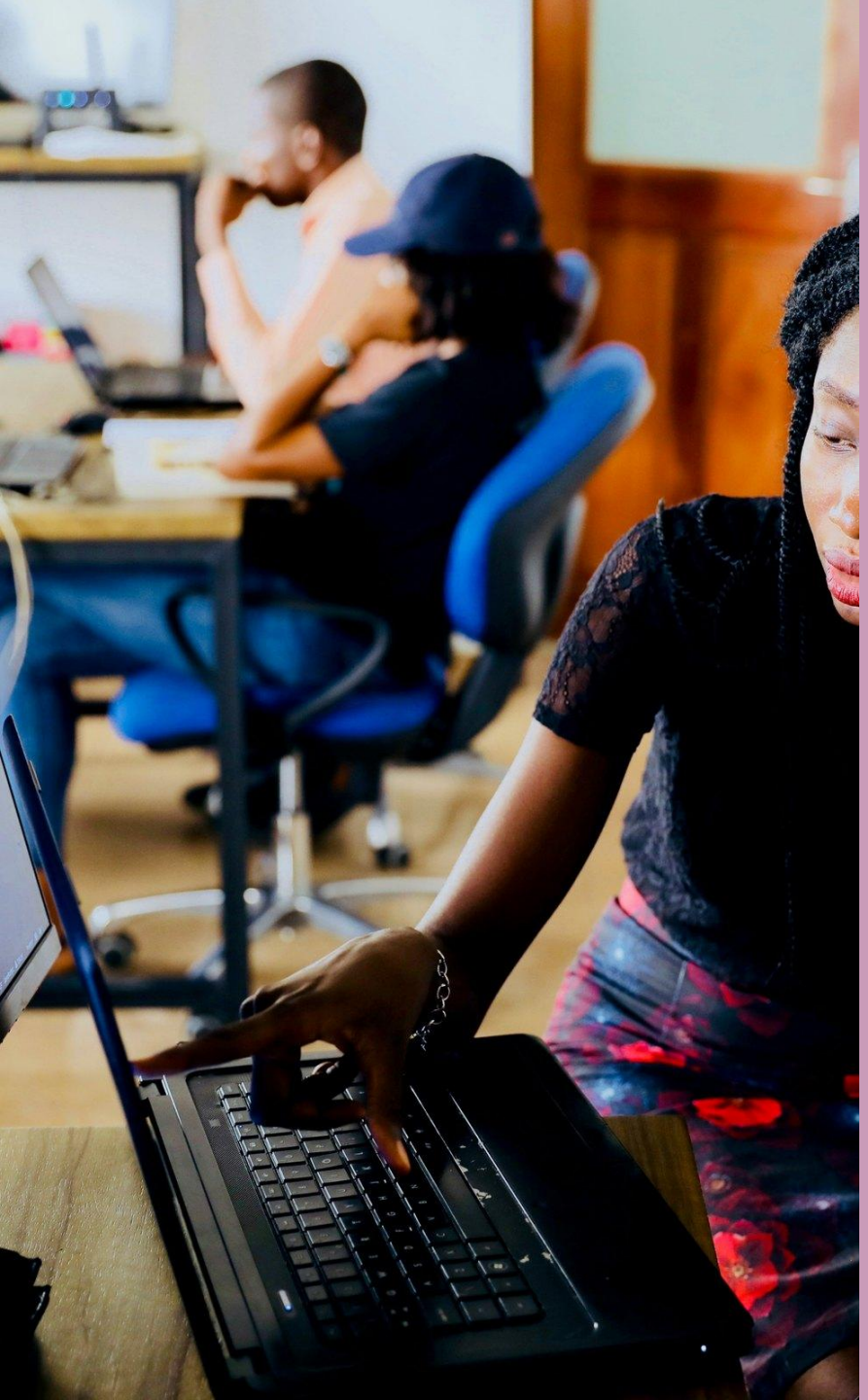
Modules checked against the published curriculum, before the last day to change them.

Reminders

Registration day, reply-by dates, and money safety before paying a deposit.

BUILT

Phase 5, built and audited. Campus and curriculum records are drafts until verified.



16 · UNIVERSITY YEARS

Every year, until the last exam

NOW

Module cards, mark calculator

How each module is marked: DP, subminimum, and the exam mark still needed. Works offline.

Free books

Open textbooks and the university library for each module, free and legal.

Plan maps and My CV

Goal maps you tick off; projects, skills and marks into a first CV (off for real learners until consent v4).

NEXT

Study coach

Reads marked tests, finds weak topics, plans around busy weeks. Phase 7, Mar 2028.

A plan that fits your life

A study week with room for classes, travel, work, family and rest. Phase 7.

Portfolio, My thread

Projects built along the way, one degree-long project. Proposed.

The aim: catch a missed DP or a wrong module before the next semester, while there is still time to fix it.

Forecasts from the learner's own marks, explained

AT SCHOOL

Where marks are heading

The pass her marks would give, and each subject's change since Grade 10.

→

Every course, forecast

Qualify now, on track, close or Plan B, each scored the university's own way.

→

What I need

The mark to aim for in each subject to reach her goal.

→

Re-check

After each new report: which courses moved up, which slipped.

AT UNIVERSITY, AND ALWAYS

The exam mark needed

From the year mark so far: DP met or not, the subminimum, what the exam must give.

It remembers

Goals, preferences and notes, kept across years. The learner sees and deletes them.

Knows when to be quiet

Messages fit her exam timetable: at most one a day, never during a paper or after 21:00.

Reads distress first

A crisis check on every message, before the AI, then a person to talk to.

How it thinks: the AI picks among 18 tools, code works out the answer, the AI explains it. Next (Phase 7): a study coach that ranks topics by the marks they cost across tests and their weight in the exam.

Built: packages/admissions (eligibility, school-record, what-i-need, recheck, mark-needed), packages/agent (tools.ts, crisis.ts), packages/messaging (cadence.ts). Phase 7: docs/plans/phase-7-proposal.md.

Connectivity and accessible learning

Light on data

Built for cheap phones: the shared code every page loads is held under 150 KB, checked on every change.

BUILT

SMS and WhatsApp

Reminders and replies without the app or a data bundle.

BUILT ON STAND-INS

Works offline

The mark calculator runs on the phone. Lost signal? It says so and tries again.

BUILT

Read aloud

Campus places and directions read out, with step-free routes.

BUILT

Free, legal books

Open textbooks and practice for school, the university library for every module.

BUILT

Accessible by design

WCAG 2.2 AA, every journey tested with axe, from a 360 px phone up, reduced motion.

BUILT

Five languages in a crisis

Crisis words recognised in isiZulu, isiXhosa, Sesotho, Setswana and Afrikaans.

BUILT

Free for learners

No fee, no adverts, nothing sold. The payers are those who gain when learners finish.

BUILT

SMS and WhatsApp run on stand-ins until a provider is chosen. Crisis phrases await first-language review.

Technical architecture

Web app

Next.js 16, installable, any browser

Phone app

Expo: Android and iOS, one codebase

SMS and WhatsApp

Reminders, replies, STOP honoured

Data, in South Africa

Postgres and file storage in AWS Cape Town (af-south-1). Row-level security. IDs only in logs. The demo: made-up learners only.

API on Vercel, Cape Town

Sign-in by phone PIN · guardian consent gate · rate limits · security headers. The app never holds a key or calls the AI.

Code decides

admissions: scores, verdicts, What I need, mark maths
catalogue: every record with source, year, checked by
messaging · documents (each mark confirmed) · planner

The AI explains

crisis check first, then the model behind an interface (Claude; Gemini on the demo only), 18 tools, the care charter. Red team: over 450 cases, 100%.

Every provider swappable

AI, SMS, WhatsApp, push, voice, CV (Mmereko), jobs, errors: each behind an interface, with a fake and contract tests.

One TypeScript monorepo · over 5 000 automated tests · guards in CI · 13 audit types per phase · E2E with axe on a production build

Live demo: six made-up learners, one journey

01	02	03	04	05	06
Thabo Grade 10	**Naledi** Grade 11	**Amahle** Grade 12	**Lerato** First year, UJ	**Sipho** Second year, Wits	**Kagiso** Final year, UCT
My subjects: swap a subject and see which courses it opens or closes.	My results, then What I need for the course she wants.	Applications and their checklists, then Funding for NSFAS.	First week: tick a step, then the campus tour.	My modules, then ask the chat how to find a place.	The campus tour, then what's coming next.

DEMO ACCOUNT · NO PASSWORD NEEDED

universityflow.tumoolo.tech/demo

Press **Try as Naledi** (or any learner): a private copy, deleted after 24 hours. Or **Sign up as a new learner** with any 080 number: the PIN shows on screen. University facts are drafts, labelled "Not yet verified". The chat is the real AI, with every safety check on.

DEMO VIDEO (88 s): https://youtu.be/MaMFUa-UEzA

65 changes shipped from 23 to 25 October

65

pull requests merged during the hackathon, each through the full test gate before it went live

Before the weekend: Phases 0 to 5 (the school years to arrival) were built and audited, September to October 2026.

The judge demo	Six made-up learners, a private copy each, a three-minute guide, live on universityflow.tumoolo.tech.
Sign up as a new learner	The real sign-up journey on made-up details: PIN and guardian link on screen.
Phase 6: semester setup	How marks work, the module outline, module cards and the mark calculator, free books.
50 TVET colleges	Every public college in Plan B, and "Is this college registered?"
My CV and plan maps	A first CV (with Mmereko, behind a switch); plan maps you pan, zoom and tick off.
Funding and polish	Ten bursaries beside NSFAS, the deadline first; clearer cards; a chat that reads as lists, not code.
Safer	Two Critical framework advisories patched; two crisis gaps closed, each with a failing test first.

From the repository history, 23 to 25 October 2026: pull requests #81 to #147, not counting automated dependency updates and one empty merge.



Built for children's data

Data stays in South Africa

Real learners' data in AWS Cape Town (af-south-1). The demo holds made-up learners only.

Guardian consent

Nothing about an under-18 is processed before a guardian's consent is recorded, in the database.

Code decides

Requirements, deadlines and scores only from checked records. The AI never works them out.

Never their password

We never log into a university system as a learner. A student number is profile data.

POPIA by design

IDs only in logs and jobs. Learners confirm every mark read from a document.

Care first

A crisis check on every message, a 100% red-team suite, a human route always.

Free for learners, paid for by those who gain

R0

for learners to use UniversityFlow. Those who lose money when a student drops out pay instead.

Free means the app. University costs stay the learner's: application and registration fees, tuition, residence, paid by the family or through NSFAS or a bursary. UniversityFlow explains them, and never pays or charges for them.

One NSFAS-funded student costs about R80 600 a year (R47.86 billion for 593 543 students in 2025). Keep one learner in 200 from dropping out, and the sponsorship for all 200 has paid for itself.



Sponsor a learner — R420 a year each

R35 a month, paid by companies (CSI and B-BBEE), bursary funders and SETAs. Anonymous impact reports.



Schools — R6 000 a year

Per fee-paying school, Grades 10 to 12. No-fee schools are free, sponsored.



Universities, from 2028 — R250 000 a year

Per university: first-year arrival, registration check, study coach. A pilot at R100 000.



Employers, from 2029 — R2 500 a month

Reach graduates who chose to share their CV (18 and over only).

R80 600 is our arithmetic on the Minister's reply to Parliament (NW874, 2026). Prices are UniversityFlow's proposal (25 Oct 2026); no payer has signed yet.

About R10.50 a learner a month to run

PER ACTIVE LEARNER, PER MONTH	
AI chat: 30 replies (Claude Sonnet 5.5, cached)	R7.50
Reading a school report (Claude Opus 5.5)	R0.35
SMS: sign-in PINs and reminders	R1.20
WhatsApp: 10 utility messages	R1.20
Hosting, per learner	R0.25
Variable cost	**R10.50**

FIXED, PER MONTH (YEAR 1)	
Hosting and database, Cape Town	R2 000
WhatsApp provider, monitoring, jobs, voice	R4 800
Legal and POPIA, fact checkers	R11 000
App stores	R150
Team: founder and a part-time developer	R45 000
Fixed cost	**about R63 000**

R24.50

margin a month on each sponsored learner (R35 less R10.50)

2 600

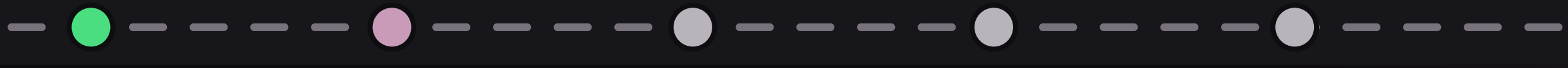
sponsored learners to break even

R1.8 m

Year 1 target: 5 000 learners (4 000 sponsored) and 20 schools, against about R1.39 m in costs

Estimates, 25 Oct 2026, at R18 to the US$. AI: Anthropic list prices with prompt caching; Haiku 5.5 for simple replies would bring the total to about R6. SMS and WhatsApp: plan §7.4. Hosting: docs/deploy/costs.md. Measured once the real AI key runs.

Timed to the first cohort: Grade 11 in 2026



NOW · OCT 2026

Built

Phases 0 to 5 and most of 6 on stand-ins, tested and audited. The judge demo is live.

MAR 2027 · MVP

Applications open

Results, applications, funding, for real learners once data, legal and accounts are in.

JUN 2027

Exams, app stores

WhatsApp, SMS and push live before the June exams.

JAN TO FEB 2028

Arrive and set up

Offers, registration, then semester setup (Phase 6).

MAR TO NOV 2028

Coach and portfolio

Study coach (Phase 7), then portfolio and career (Phase 9).

PARTNER GATES · EACH ONLY WITH A SIGNED AGREEMENT

H1 2027: apply with UniversityFlow at one university (UJ first). H2 Jan 2028: official NSC results with the DBE. H3 2028: NSFAS and bursary funders accept pre-filled applications.



THE DESTINATION

A learner who finishes

Our ask: help us open UniversityFlow to real learners by March 2027.

- People to verify the university data (86 programmes, six universities)
- A legal review of consent for under-18s (POPIA)
- An SMS and WhatsApp provider
- Hosting in AWS Cape Town (af-south-1) for learners' data
- A first university partner, and bursary funders

UniversityFlow *From Grade 10 to graduation.*

APPENDIX

Sources

1. DBE, 2025 NSC Examination Report (Tables 1, 2, 2.1, 3, 6.3.8), 2026. education.gov.za
2. DBE, 2024 NSC Examination Report (Tables 1, 6.1.4, 6.3.3; Figure 2), 2025. education.gov.za
3. DBE, NSC Examination Report 2021 (Tables 2, 6.2.1, 6.3.9), 2022. education.gov.za
4. DHET, 2000 to 2019 First Time Entering Undergraduate Cohort Studies for Public HEIs, 22 Feb 2022, Table 28. dhet.gov.za
5. Minister of Higher Education, written reply to question NW874 (NSFAS funding 2023 to 2025), 2026. pmg.org.za/committee-question/37471
6. NSFAS household income criteria, as published by Rhodes University (ru.ac.za/registrar/nsfas); NSFAS's own site could not be reached
7. Stats SA, Quarterly Labour Force Survey Q2 2026, media release and presentation, 11 Aug 2026. statssa.gov.za
8. UniversityFlow plan (planning.md §0, §1, §2, §7, §10, §11, §19) and status (status.md, 24 Oct 2026), in the repository
9. Built behaviour: packages/messaging (cadence.ts), packages/agent (crisis.ts, charter.ts, system-prompt.ts), packages/content (messages.ts)
10. Screens: the live judge demo, universityflow.tumoolo.tech, made-up learners, captured 25 Oct 2026
11. Costs: Anthropic list prices (Oct 2026); docs/deploy/costs.md (hosting, checked 23 Oct 2026); plan §7.4 (SMS, WhatsApp). Estimates at R18 to the US dollar
12. Full list with exact URLs, numbers and image credits: pitch/README.md